2024, Mar. 2024 Release LOC:Acute Pediatric Dehydration or Gastroenteritis

Utilization Benchmarks: *Multiple options available, see table below for options*

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Overview

Instruction:
This subset is for the management of dehydration or uncomplicated infectious gastroenteritis and excludes hemolytic uremic syndrome. For electrolyte abnormalities or clinical findings that do not meet criteria in dehydration or gastroenteritis, refer to the appropriate condition specific subset.

Introduction:
Infective gastroenteritis is the most common cause of diarrhea and vomiting in young children. Most children experience 1-2 episodes per year. In developed countries, most cases are of viral etiology. Since the introduction of the rotavirus vaccine, the disease burden has decreased substantially. Gastroenteritis is characterized by the sudden onset of diarrhea, with or without vomiting, fever, or abdominal pain. Children with diarrhea may become dehydrated and most can be managed in the home or community setting. Inpatient admission may be required for moderate to severe cases of dehydration (Florez et al., Curr Infect Dis Rep 2020, 22: 4).

Evaluation and Treatment:
Gastroenteritis should be suspected when there is a sudden change in stool consistency or frequency, or when a child is vomiting.
The following children are at increased risk for dehydration:

- Children under 1 year
- Children who have passed more than 3 diarrhea stools over the past 72 hours
- Children who have vomited more than twice in the previous 24 hours

While a causative pathogen is rarely identified in gastroenteritis, stool sampling is recommended for those who have recently traveled abroad, have had diarrhea for a period longer than seven days, or when there is uncertainty about the diagnosis of gastroenteritis. Severity of dehydration should be determined in children with gastroenteritis by assessing the following: weight loss (if attainable), general appearance, mental status, urine output, peripheral perfusion, vital signs, and skin turgor. Mild dehydration can be managed in the home or community setting with oral replacement therapy (ORT). Children requiring hospital admission for moderate to severe dehydration may require laboratory evaluation of sodium, potassium, blood urea nitrogen (BUN), creatinine, and glucose concentrations. Intravenous (IV) fluids should only be considered when a child is, unable to tolerate ORT (orally or via a nasogastric tube), has altered mental status, difficulty breathing, or is exhibiting signs of shock (Florez et al., Curr Infect Dis Rep 2020, 22: 4).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence

(NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the Cochrane Handbook. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources (Florez et al., Curr Infect Dis Rep 2020, 22: 4; Acute Gastroenteritis Guideline Team, Cincinnati Children's Hospital Medical Center: Evidence-based care guideline for prevention and management of acute gastroenteritis in children aged 2 months to 18 years. 2011; Colletti et al., J Emerg Med 2010, 38: 686-98; National Guideline, Guideline for the prevention and control of norovirus gastroenteritis outbreaks in healthcare settings; Khanna et al., BMJ 2009, 338: b1350; Guarino et al., J Pediatr Gastroenterol Nutr 2008, 46 Suppl 2: S81-122).

(Excludes PO medications unless noted)
Select Day, One:
- **Episode Day 1, One:**
  (Symptom or finding within 24h)
  - **OBSERVATION, Both:** [1]
    - Age > 1 to < 18 yrs and persistent symptoms post initial treatment and, **One:**
      - Symptomatic [2, 3, 4, 5]
    - Intervention, ≥ **One:** [6, 7, 8]
      - Oral rehydration therapy ≤ 2d [9]
      - Antiemetic ≥ 3x/24h (includes PO or PR) [10]
      - Serotonin antagonist ≥ 2x/24h (includes PO) [11]
      - IV fluid [12, 13, 14]
  - **ACUTE, Both:** [15]
    - Finding, ≥ **One:** [16]
      - Age ≤ 1 yr and symptomatic [2, 4, 5]
      - Listlessness or lethargy [17]
      - Chloride < lower limit of normal or > ULN [18]
      - Potassium 2.5-3.2 mEq/L(2.5-3.2 mmol/L) and no ECG changes [19, 20, 21]
      - Sodium 146-158 mEq/L(146-158 mmol/L) [22]
    - Intervention, ≥ **One:** [23, 24]
      - Oral rehydration therapy ≤ 2d [9]
      - Antiemetic ≥ 3x/24h (includes PO or PR) [10]
      - Serotonin antagonist ≥ 2x/24h (includes PO) [11]
      - Potassium repletion (includes PO)
      - IV fluid [12, 13, 14]
      - Rehydration therapy via nasogastric tube [25]
  - **CRITICAL, ≥ One:** [26]
    - Hypernatremia **(use Electrolyte and Mineral Imbalance subset)**
    - Bicarbonate ≤ 17 mmol/L and symptomatic [27, 28]
    - Coma, stupor, or somnolence [29]
    - Hypokalemia, **All:** [21, 30, 31]
      - Potassium < 3.2 mEq/L(3.2 mmol/L)
      - Electrocardiogram (ECG) findings, ≥ **One:** [21]
        - Multifocal PVCs
        - PVCs > 6/min
        - R on T phenomenon [32]
        - T and U wave fusion [33]
        - U wave [34]
      - Potassium repletion
    - Hemodynamic instability, **Both:**
      - Finding, ≥ **Two:**
        - Heart rate, **One:** [35]
          - > 180/min, sustained (age ≤ 12 mos) [36]
          - > 160/min, sustained (age > 12 mos to 4 yrs) [36]
          - > 150/min, sustained (age > 4 yrs to 8 yrs) [36]
          - > 140/min, sustained (age > 8 yrs to 15 yrs) [36]
          - > 130/min, sustained (age > 15 yrs to < 18 yrs) [36]
        - Systolic BP, **One:**
          - < 70 mmHg (age ≤ 12 mos)
          - < 80 mmHg (age > 1 to < 5 yrs)
          - < 90 mmHg (age 5 to < 18 yrs)
          - Labile [37]
        - MAP, **One:** [38]

- < 55 mmHg (≤ 12 mos)
- < 65 mmHg (> 1 to < 18 yrs)
- IV fluid resuscitation ≤ 2d [39]
- Mechanical ventilation
- NIPPV and, ≥ **One:** [40]
  - Arterial Po_2 ≤ 39 mmHg(5.2 kPa)
  - Arterial or venous Pco_2 ≥ 60 mmHg(8.0 kPa)
  - Arterial or venous pH ≤ 7.24
- **Episode Day 2, One:**
  - **OBSERVATION, One:**
    - **Responder**, discharge expected today if clinically stable last 12h, **All:** [41]
      - Tolerating PO [42]
      - Vomiting or diarrhea improving or not applicable
      - Urine output improving or within acceptable limits
      - No complication or active comorbidity relevant to this episode of care [43]
    - **Partial responder**, not clinically stable for discharge and requires continued stay, **Both:**
      - Age > 1 to < 18 yrs and persistent symptoms [2, 3, 4, 5]
      - Intervention, ≥ **One:** [6, 23, 24]
        - Oral rehydration therapy ≤ 2d [9]
        - Antiemetic ≥ 3x/24h (includes PO or PR) [10]
        - Serotonin antagonist ≥ 2x/24h (includes PO) [11]
        - IV fluid [12, 13, 14]
  - **ACUTE, One:**
    - **Responder**, discharge expected today if clinically stable last 24h, **All:** [41]
      - Tolerating PO [42]
      - Vomiting or diarrhea improving or not applicable
      - Laboratory values within acceptable limits [44]
      - Mental status within normal limits or returned to baseline
      - Urine output improving or within acceptable limits
      - No complication or active comorbidity relevant to this episode of care [43]
    - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
      - Persistent symptoms, **Both:**
        - Finding, ≥ **One:**
          - Age ≤ 1 yr and persistent symptoms [4, 5]
          - Mental status change (excludes coma, stupor, or somnolence) [17, 29]
          - Chloride < lower limit of normal or > ULN [18]
          - Potassium 2.5-3.2 mEq/L(2.5-3.2 mmol/L) and no ECG changes [19, 20, 21]
          - Sodium 146-158 mEq/L(146-158 mmol/L) [22]
        - Intervention, ≥ **One:** [23, 24]
          - Oral rehydration therapy ≤ 2d [9]
          - Antiemetic ≥ 3x/24h (includes PO or PR) [10]
          - Serotonin antagonist ≥ 2x/24h (includes PO) [11]
          - Potassium repletion (includes PO)
          - IV fluid [12, 13, 14]
          - Rehydration therapy via nasogastric tube [25]
      - Post Critical care ≤ 24h
  - **INTERMEDIATE, ≥ One:** [45]
    - NIPPV [40]
    - Oxygen 35-39%(0.35-0.39) [46]
  - **CRITICAL, ≥ One:**
    - Hypernatremia **(use Electrolyte and Mineral Imbalance subset)**
    - Bicarbonate ≤ 17 mmol/L and symptomatic [27, 28]
    - Coma, stupor, or somnolence [29]

- Hypokalemia, **All:** [21, 30, 31]
  - Potassium < 3.2 mEq/L(3.2 mmol/L)
  - Electrocardiogram (ECG) findings, ≥ **One:** [21]
    - Multifocal PVCs
    - PVCs > 6/min
    - R on T phenomenon [32]
    - T and U wave fusion [33]
    - U wave [34]
  - Potassium repletion
- Hemodynamic instability, **Both:**
  - Finding, ≥ **Two:**
    - Heart rate, **One:** [35]
      - > 180/min, sustained (age ≤ 12 mos) [36]
      - > 160/min, sustained (age > 12 mos to 4 yrs) [36]
      - > 150/min, sustained (age > 4 yrs to 8 yrs) [36]
      - > 140/min, sustained (age > 8 yrs to 15 yrs) [36]
      - > 130/min, sustained (age > 15 yrs to < 18 yrs) [36]
    - Systolic BP, **One:**
      - < 70 mmHg (age ≤ 12 mos)
      - < 80 mmHg (age > 1 to < 5 yrs)
      - < 90 mmHg (age 5 to < 18 yrs)
      - Labile [37]
    - MAP, **One:** [38]
      - < 55 mmHg (≤ 12 mos)
      - < 65 mmHg (> 1 to < 18 yrs)
  - IV fluid resuscitation ≤ 2d [39]
- Mechanical ventilation
- NIPPV and, ≥ **One:** [40]
  - FiO_2 ≥ 40%(0.40)
  - Respiratory intervention every 1-2h [47]
- Oxygen ≥ 40%(0.40) [46]

Episode Day 3, One:

- **OBSERVATION, One:** [1]
  - **Responder**, discharge expected today if clinically stable last 12h, **All:** [41]
    - Tolerating PO [42]
    - Vomiting or diarrhea improving or not applicable
    - Urine output improving or within acceptable limits
    - No complication or active comorbidity relevant to this episode of care [43]
  - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
    - For a condition other than dehydration or gastroenteritis use appropriate subset (Episode Day 1) based on clinical findings
    - Refer for secondary review
- **ACUTE, One:**
  - **Responder,** discharge expected today if clinically stable last 24h, **All:** [41]
    - Tolerating PO [42]
    - Vomiting or diarrhea improving or not applicable
    - Complication or comorbidity, **One:** [43]
      - No complication or active comorbidity relevant to this episode of care [43]
      - Complication or comorbidity and clinically stable for discharge, ≥ **One:**
        - Laboratory values within acceptable limits [44]
        - Mental status within normal limits or returned to baseline
        - Urine output improving or within acceptable limits

- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
  - **Use Extended Stay subset**
  - Refer for secondary review
- **INTERMEDIATE**, **One:**
  - **Non-responder**, Not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**
    - **Use Extended Stay subset**
    - Refer for secondary review
- **CRITICAL**, **One:**
  - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
    - **Use Extended Stay subset**
    - Refer for secondary review

Discharge, Level of Care, One: [48]

- **HOME, Both:**
  - Level of care appropriateness, **Both:**
    - Home environment safe and accessible [49]
    - Patient or caregiver demonstrates ability to manage care
  - Dehydration or gastroenteritis discharge planning, **All:**
    - Provide comprehensive written discharge and teaching instructions [50]
    - Education on the importance of signs and symptoms of fluid loss and how to replace fluids
    - Gastroenteritis discharge education [51]
    - Medication reconciliation [52]
    - Follow-up care planned [53]
    - Identify and address transportation needs
- **HOME CARE, Both:**
  - Level of care appropriateness, **All:**
    - Home environment safe and accessible [49]
    - Patient or caregiver willing and able to learn care
    - Outpatient management contraindicated or unavailable [54, 55]
    - Skilled services, ≥ **One:** [56]
      - Skilled nursing [57]
      - Skilled therapy, PT, OT, or ST
      - Behavioral health [58]
  - Dehydration or gastroenteritis discharge planning, **All:**
    - Provide comprehensive written discharge and teaching instructions [50]
    - Education on the importance of signs and symptoms of fluid loss and how to replace fluids
    - Gastroenteritis discharge education [51]
    - Medication reconciliation [52]
    - Follow-up care planned [53]
    - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
  - Level of care appropriateness, **One:**
    - Support systems available [59]
  - Skilled treatment, ≥ **One:**
    - Clinical assessment [60]
    - Individual, group, or family counseling
    - Psychiatric, substance, or medication evaluation [61]
- **SKILLED MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk

- Treatment precluded in a lower level
- Services, ≥ **One:**
  - Medical and skilled nursing interventions or assessment 1-2x/24h [57]
  - Therapy, **All:**
    - Goal directed therapy with rehabilitation potential [62, 63]
    - Functional impairments requiring at least supervision [64]
    - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
- Complete prior to facility transfer, **All:**
  - Medication reconciliation [52]
  - Obtain and complete forms for facility
  - Obtain discharge summary and transmit to facility and medical practitioner
  - Arrange transportation

- **SUBACUTE MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 4h/24h [57]
      - Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [62, 63]
        - ≥ 2 functional impairments requiring at least minimum assistance [64]
        - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [52]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **SUBACUTE REHAB, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Treatment precluded in a lower level
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential [62, 63]
      - ≥ 2 functional impairments requiring at least minimum assistance [64]
      - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [52]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **ACUTE REHAB, Both:**
  - Level of care appropriateness, **Both:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential [62, 63]
      - ≥ 2 functional impairments requiring at least minimum assistance [64]
      - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines [65]
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [52]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**

- Level of care appropriateness, **Both:**
  - Treatment precluded in a lower level
  - In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:**
    - Primary condition or illness and treatment of active comorbid condition(s) [66]
    - Ventilator dependent ≥ 6h/d and weaning planned [67, 68]
    - Acute and comorbid conditions requiring prolonged hospitalization [69]
- Complete prior to facility transfer, **All:**
  - Medication reconciliation [52]
  - Obtain and complete forms for facility
  - Obtain discharge summary and transmit to facility and medical practitioner
  - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
DEHYDRATION	**55%**	**1.9**	**InterQual**
GASTROENTERITIS	**48%**	**2.1**	**InterQual**

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:

1:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

2:

Instruction: Measuring urine output in newborns, infants, or patients who are unable to use a bedpan or other measuring device can be done via straight catheterization, or alternatively, by weighing diapers. Weight in grams can be translated to milliliters (1 gram = 1 mL) (Fuhrman, Pediatr Crit Care Med 2019, 20: 381-2).

3:

Weight loss, although the most accurate measurement of dehydration, can be difficult to calculate precisely if weight prior to illness is not readily available (Burgunder: The Harriet Lane Handbook, 22 edn. 2020:261-82.e2).

4:

Symptoms of dehydration include:

- Absent tears
- Delayed capillary refill (> 2 seconds)
- Dry mucus membranes
- Oliguria (urine output less than 1 mL/kg/h or 180 mL/m^2/24h)
- Persistent diarrhea
- Persistent vomiting after antiemetic
- Sunken eyes

5:

Symptoms of dehydration include (Colletti et al., J Emerg Med 2010, 38: 686-98):

- Absent tears
- Delayed capillary refill (> 2 seconds)
- Dry mucus membranes
- Oliguria (urine output less than 1 mL/kg/h or 180 mL/m^2/24h)
- Persistent diarrhea
- Persistent vomiting after antiemetic
- Sunken eyes

6:

Reintroducing nutrition is an important component of the treatment of gastroenteritis. The presence of nutrients in the digestive tract promotes mucosal health and absorptive function. Following adequate rehydration, diet should be advanced to include milk and the child's usual solid foods. Fruit juices and carbonated drinks have the potential to prolong the duration of diarrhea and should be avoided until diarrhea ceases (Shane et al., Clin Infect Dis 2017, 65: e45-e80).

7:

Best practice is to replace fluids over 24h utilizing 3 rehydration phases (Burgunder: The Harriet Lane Handbook, 22 edn. 2020:261-82.e2):

- Initial stabilization (rapid fluid replacement)
- Maintenance volume and deficit repletion (maintain fluid gain and help with ongoing losses)
- Recovery (ongoing losses should be managed)

8:

Probiotics are found to be effective in slowing diarrhea output and managing frequency of stools. Initiating appropriate doses of probiotics at the first sign of symptoms is an effective adjunct to treatment for diarrhea (Brady, Pediatr Emerg Med Pract 2018, 15: 1-25; Feizizadeh et al., Pediatrics 2014, 134: e176-91; Szajewska et al., Aliment Pharmacol Ther 2013, 38: 467-76).

9:

This criteria point includes hydration via enteral feeding tubes.

10:

Antiemetics are used to treat nausea and vomiting in children and providers should be aware of the risk of extrapyramidal reactions with administration (Colletti et al., J Emerg Med 2010, 38: 686-98).

11:

Serotonin antagonists are a class of antiemetic used to reduce nausea and vomiting. Ondansetron is the most common serotonin antagonist used in children (Fedorowicz et al., Cochrane Database Syst Rev 2011; (9): CD005506; Colletti et al., J Emerg Med 2010, 38: 686-98).

12:

To allow for the advancement of feedings as intravenous (IV) fluids are weaned, the IV fluid rates are ≥ 50%(0.50) of the maintenance fluid requirement. These rates can be calculated using the following (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020):

Weight ≤ 10 *kg* = ≥ 50 *mL/kg*/24h

- Hourly rate = 2.08 (*mL/h*) x kg

Weight > 10-25 *kg* = ≥ 40 *mL/kg*/24h

- Hourly rate = 1.67 (*mL/h*) x kg

Weight > 25-60 *kg* = ≥ 30 *mL/kg*/24h

- Hourly rate = 1.25 (*mL/h*) x kg

Example: An 11 *kg* child is required to receive a minimum of 40 *mL/kg*/24h to meet this criteria point. Multiply 1.67 (*mL*/h) x 11 (child's weight in *kg*) = 18.37 *mL*/h. This is the minimum fluid replacement rate for this child.

13:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

14:

Isotonic solution is the most recommended intravenous fluid for hospitalized children between the ages of 28 days and 18 years by the American Association of Pediatrics (Burgunder: The Harriet Lane Handbook, 22 edn. 2020:261-82.e2).

15:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

16:

Children that require a specialized nutritional program may need to be admitted for dehydration due to the specific challenges of rehydration and maintaining adequate nutrition (Joshua Guttman, In: Rosen's Emergency Medicine, 9 edn. 2018:230-41.e1).

17:

Electrolyte laboratory values should be obtained when a patient presents with any of the following findings (Colletti et al., J Emerg Med 2010, 38: 686-98):

- Altered mental status
- Moderate to severe dehydration
- Symptoms of hypokalemia or hypernatremia

18:

Instruction: This line of criteria is intended for the patient who has a chloride level that is either below the lower limit of normal or above the upper limit of normal based on institutional laboratory reference ranges for normal values.

19:

In patients with no underlying heart disease, occasional, nonconsecutive premature ventricular contractions (PVCs) less than 6/min are common and not considered a concerning finding. Three or more consecutive PVCs at a rate of 100 bpm or more that terminate spontaneously within 30 seconds are considered nonsustained ventricular tachycardia (Marine, Card Electrophysiol Clin 2016, 8: 525-43; Buxton et al., Circulation 2006, 114: 2534-70).

20:

Following an initial 12-lead electrocardiogram (ECG) demonstrating no abnormal findings in the setting of hyperkalemia or hypokalemia, continued monitoring may be accomplished by telemetry (Bridgeman et al., Nephrol Dial Transplant 2019, 34: iii45-iii50).

21:

Hypokalemia (Sandau et al., Circulation 2017, 136: e273-e344; Zieg et al., Acta Paediatr 2016, 105: 762-72)

22:

Hypernatremia (Muhsin and Mount, Best Pract Res Clin Endocrinol Metab 2016, 30: 189-203)

23:

Selected probiotics, such as *Lactobacillus rhamnosus* GG (LGG), have been shown to decrease the duration of diarrhea and should be considered as an adjunctive therapy for patients with gastroenteritis (Shane et al., Clin Infect Dis 2017, 65: e45-e80; Freedman et al., PLoS One 2015; 10(6): e0128754; Allen et al., Cochrane Database Syst Rev 2010; (11): CD003048).

24:

The use of anti-diarrheals in pediatric patients for the management of gastroenteritis is not recommended. However, the use of ondansetron may decrease vomiting and hospitalization rates in patients who require intravenous or nasogastric tube fluid replacement. Caution should be used in administering these agents as they may prolong the duration of diarrhea and can cause fatal arrhythmias (Shane et al., Clin Infect Dis 2017, 65: e45-e80; Freedman et al., PLoS One 2015; 10(6): e0128754; Fedorowicz et al., Cochrane Database Syst Rev 2011; (9): CD005506).

25:

Nasogastric tube placement may be required if the child cannot take fluids by mouth after antiemetics, has poor peripheral IV access, or to avoid IV placement (Brady, Pediatr Emerg Med Pract 2018, 15: 1-25; MacGillivray et al., Cochrane Database of Systematic Reviews 2013:).

26:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a

variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

27:

A child who presents with clinical symptoms of dehydration, such as, deep sunken eyes, listlessness, or poor skin turgor (greater than 2 seconds) and a serum bicarbonate concentration of 17mmol/L or less should be managed at the Critical level of care (Colletti et al., J Emerg Med 2010, 38: 686-98; Vega and Avner, Pediatr Emerg Care 1997, 13: 179-82).

28:

Symptoms include:

- Deep sunken eyes
- Listlessness
- Poor skin turgor > 2 sec

29:

For patients at the Acute level of care, mental status change may include disorientation, agitation, increasing irritability, listlessness, or lethargy. The inability to soothe an infant, abnormal cry (e.g., weak, high-pitched), or decreased social interaction with parents or caregivers in children 2 years old or younger, may also qualify as a mental status change (Keith Kleinman, In: Harriet Lane Handbook, 22nd edn. 2021).
Instruction: Patients with acute coma, stupor, or somnolence should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma or stupor.

30:

When a patient with hypokalemia is not responding to potassium repletion, the magnesium level should be checked since hypomagnesemia could be contributing to the refractory hypokalemia.

31:

Hypokalemia rarely occurs in isolation and is usually secondary to a disease process or overuse of diuretics. Criteria should be reviewed for the underlying cause of the hypokalemia (e.g., diabetic ketoacidosis, gastroenteritis) to ensure the patient is managed appropriately.

32:

The "R on T phenomenon" is the superimposition of an ectopic beat on the T wave of a preceding beat. Early observations suggest that R on T is likely to initiate sustained ventricular tachyarrhythmias.

33:

When T and U waves fuse, they form a TU fusion wave on an electrocardiogram. Severe hypokalemia may be associated with a large U wave created by T and U wave fusion (Zieg et al., Acta Paediatr 2016, 105: 762-72; Levis, Perm J 2012, 16: 53).

34:

The U wave occasionally follows the T wave on an electrocardiogram. A prominent upright U wave is associated with hypokalemia. When the U wave exceeds the T wave amplitude, the serum potassium level is less than 3 mEq/L. Severe hypokalemia may be associated with a giant U wave created by T and U wave fusion (Sandau et al., Circulation 2017, 136: e273-e344; Zieg et al., Acta Paediatr 2016, 105: 762-72).

35:

In the pediatric patient, hemodynamic instability could be assessed by an elevated heart rate (HR) above the 95^{th} percentile norms based on specific age groups. A combination of different parameters and their trends could also be utilized to ascertain impending clinical deterioration more accurately in these patients to allow for closer monitoring in the critical care setting, including respiratory rate, mental status, and capillary refill time, amongst others. Other factors that could impact HR in the pediatric population include pain, anxiety, medications, and/or fever. Although no specific consensus has been reached on HR thresholds that signify unstable hemodynamics by various pediatric societies, systemic reviews and observational studies have compared HR data to textbook references such as the Pediatric Advance Life Support (PALS). Eytan et al., in 2017, studied HR in pediatric patients who were critically ill in the pediatric intensive care unit (PICU) and developed percentiles of heart rates categorized into the same age groups used in previous studies, including PALS, with the goal of comparing their observed norms, especially between the 5^{th} and 95^{th} percentiles (Eytan et al., Frontiers in Pediatrics 2017, 5). Using their HR data above the 95^{th} percentile, HR thresholds were utilized to identify the pediatric patient who may require immediate or urgent intervention and rapidly treat unstable hemodynamics to prevent a potentially serious event.

36:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).
Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

37:

Blood pressure is considered labile when it is extremely variable and difficult to control.

38:

In pediatric patients, hemodynamic instability may be defined as a mean arterial pressure (MAP) less than 55 mmHg in infants up to 1 year of age and less than 65 mmHg in patients over 1 year of age (Matics and Sanchez-Pinto, JAMA Pediatr 2017, 171: e172352).

39:

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, N Engl J Med 2018, 378: 862-3).
Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zengin et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

40:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be

delivered at concentrations approximating 100%(1.0).
The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.
Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

41:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

42:

Oral intake should be greater than or equal to the patient's maintenance fluid requirements or within parameters which the medical practitioner determines are acceptable. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid requirements (*mL/kg*/d) can be calculated based on current body weight (*kg*):

- 100 *mL/kg*/d for first 10 *kg*
- 50 *mL/kg*/d for second 10 *kg*
- 20 *mL/kg*/d for each additional *kg* in weight

43:

Relevant complications or comorbidities are conditions that actively require management during an inpatient stay.

44:

Lab values include:

- Electrolytes
- Chloride

45:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

46:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air [room air: 21%(0.21) FIO_2] to correct hypoxia. For pediatric patients, the selection of an oxygen delivery system needs to take into consideration the patient's size, needs, and therapeutic goals. The oxygen concentration or percentage (FIO_2) delivered varies with the manufacturer's design, delivery device, patient's minute ventilation (dependent on the respiratory rate and tidal volume), and oxygen flow rate. In general, patients with higher minute ventilation (common in most sick patients and those with respiratory disease) require considerably higher flow rates to correct hypoxia. The following are examples of oxygen delivery systems commonly used for pediatric patients with the percentage range of FIO_2 optimally delivered:
Low-flow systems

- Nasal cannula / Nasopharyngeal catheters 24-35%(*0.24-0.35*) FIO_2

Reservoir systems (For neonates, infants, and young children, the maximum FIO_2 achievable is not well documented)

- Simple oxygen masks 35-50%(*0.35-0.50*) FIO_2

- Partial-rebreathing masks 40-60%(*0.40-0.60*) FIO_2
- Non-rebreathing masks > 60%(*0.60*) FIO_2

High-flow systems

- Air-entrainment masks / nebulizers 24-40%(*0.24-0.40*) FIO_2
- Heated, humidified, high flow oxygen concentrating nasal cannula 21-100%(*0.21-1.0*) FIO_2

Enclosure systems

- Oxygen hoods / tents 21-100%(*0.21-1.0*) FIO_2

For children less than 5 years old, the oxygen regimen may need to be adjusted to a lesser concentration and flow rate, while still requiring acute level of care due to the patient's size, metabolic needs, and vital sign instability.

47:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

48:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

49:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

50:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

51:

Discharge teaching for gastroenteritis should include information about controlling disease transmission. Children may return to school or daycare when the patient is afebrile, has not vomited in 24 hours, when stools are more formed and can be adequately contained, and when there is assurance the facility adheres to appropriate hand washing policies. Caregivers should be instructed to monitor the patient for signs and symptoms of progressive dehydration and receive information on what to do if dehydration does progress. Additionally, guidelines recommend immunization against rotavirus for infants (Acute Gastroenteritis Guideline Team, Cincinnati Children's Hospital Medical Center: Evidence-based care guideline for prevention and management of acute gastroenteritis in children aged 2 months to 18 years. 2011).

52:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

53:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

54:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

55:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

56:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen

- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

57:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

58:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.
Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.
Evaluation and Treatment
Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

59:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

60:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures

- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

61:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

62:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

63:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

64:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.
The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

65:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

66:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

67:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

68:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

69:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.